

SprintFlow:

Agile Project Management Information System

Abdul Basit (23K-0526)
Alishba Rasheed (23K-0756)
Taaha Khan (23K-0583)

Advisor: Abdul-Rahman Mahmood

Date: 16 February 2026

Contents

Contents	2
1. Overview	3
2. Project Detail	3
2.1 Problem or Challenge	3
2.2 Project Vision	4
2.3 Project Feasibility Study	4
2.3.1 Technical Feasibility	4
2.3.2 Operational Feasibility	4
2.3.3 Economic Feasibility	4
2.4 Project Goals	5
Functional Goals	5
Technological Goals	5
Quality Goals	5
Organizational Goals	5
Other Goals	5
Constraints	5
2.5 Project Abstract	5
2.6 Project Scope	6
Included	6
Excluded	6
2.7 Project Team	7
2.8 Existing & Related Work	7
2.9 Proposed Solution & Architecture	8
2.10 Use Case Model	8
Actors	8
Major Use Cases	8
2.11 Feature List (Core Domain Features)	8
2.12 Software Development Methodology	9
2.13 Proposed Infrastructure Platform and Tools	9
2.14 Project Timeline	9
2.15 Mockups (Optional)	10
2.16 References	10

1. Overview

Software development projects often involve multiple team members, tasks, deadlines, and dependencies. Managing these elements manually or through basic tools results in inefficiency, poor coordination, and missed deadlines. The motivation behind SprintFlow is to provide a centralized system for managing software projects using Agile principles.

The primary customers are software development teams, project managers, organizations, and student groups working on collaborative projects.

SprintFlow will deliver a new web-based Management Information System (MIS) designed for project planning, task tracking, sprint management, and performance monitoring. It is not an extension of an existing system but a newly developed application inspired by modern tools.

The project is expected to be completed within the academic term duration and will not depend on other projects for its completion.

2. Project Detail

2.1 Problem or Challenge

In Pakistan and similar developing regions, many organizations and student teams lack access to affordable and structured project management tools. Existing international tools are often costly, complex, or not tailored to local needs.

As a result:

- Teams struggle with task coordination
- Progress tracking is inefficient
- Communication gaps arise
- Deadlines are frequently missed
- Accountability is unclear

This creates a need for a locally developed, easy-to-use, and cost-effective project management system.

2.2 Project Vision

The vision of SprintFlow is to create a comprehensive, user-friendly Agile Project Management Information System that enables teams to plan, execute, and monitor projects efficiently.

The system aims to become a centralized platform where stakeholders can track project progress, manage tasks, collaborate in real time, and make informed decisions based on analytics. Success will be achieved when teams can complete projects more efficiently with improved transparency and coordination.

2.3 Project Feasibility Study

2.3.1 Technical Feasibility

The project is technically feasible because it uses widely available web technologies and tools. The development team possesses the necessary skills in web development, databases, and software engineering practices.

Required infrastructure such as computers, internet connectivity, and development software is readily available.

2.3.2 Operational Feasibility

The system will be user-friendly and designed to require minimal training. It will integrate naturally into existing workflows of teams and organizations, making adoption practical and beneficial.

2.3.3 Economic Feasibility

Development costs are minimal since open-source technologies will be used. No expensive hardware or proprietary software licenses are required. Therefore, the project is economically viable.

2.4 Project Goals

Functional Goals

- Manage projects, tasks, issues, and sprints
- Provide collaboration features
- Generate reports and analytics

Technological Goals

- Develop a scalable web application
- Implement secure access control

Quality Goals

- Reliability
- Performance
- Maintainability

Organizational Goals

- Improve team coordination
- Support decision-making

Other Goals

- Usability
- Accessibility
- Portability

Constraints

- Limited development time
- Academic project scope
- Available hardware/software resources

2.5 Project Abstract

SprintFlow is a web-based Agile Project Management Information System designed to support planning, execution, monitoring, and control of software projects. The system provides tools for

project creation, team management, task assignment, sprint planning, issue tracking, and performance reporting within a centralized platform.

Inspired by professional tools such as Jira, Trello, and Asana, SprintFlow aims to deliver similar functionality in a simplified and cost-effective form suitable for local organizations and student teams.

The application will support Agile methodologies including Scrum and Kanban. Users will be able to manage product backlogs, organize sprints, visualize tasks using boards, track issues, collaborate through comments, and monitor progress through dashboards and reports.

The system will also implement role-based access control to ensure security and accountability. By integrating project planning, workflow management, communication, and analytics into one platform, SprintFlow aims to enhance productivity, transparency, and timely project completion.

2.6 Project Scope

Included

- Project management features
- Task and issue tracking
- Sprint and backlog management
- Collaboration tools
- Reporting and analytics

Excluded

- Mobile application (web only)
 - Integration with external enterprise systems
 - Financial management features
-

2.7 Project Team

Organization	Name	Availability	Comment
FAST-NU	Abdul Basit	Full-time	Team Leader
FAST-NU	Alishba Rasheed	Part-time	Developer
FAST-NU	Taaha Khan	Part-time	Developer

2.8 Existing & Related Work

Tool	Key Features	Limitations	Uniqueness of SprintFlow
Jira	Advanced tracking	Complex & costly	Simplified MIS version
Trello	Kanban boards	Limited analytics	Full lifecycle management
Asana	Task coordination	Paid features	Free academic solution

2.9 Proposed Solution & Architecture

SprintFlow will use a layered web architecture:

Presentation Layer: User Interface

Application Layer: Business Logic

Data Layer: Database

Users interact through a browser → requests processed by backend → data stored/retrieved from a database.

2.10 Use Case Model

Actors

- Administrator
- Project Manager
- Team Member

Major Use Cases

- Manage projects
 - Manage tasks
 - Plan sprints
 - Track issues
 - Generate reports
 - Collaborate with team
-

2.11 Feature List (Core Domain Features)

1. Project creation and management
2. Team assignment to projects
3. Task creation and assignment
4. Issue reporting and tracking
5. Sprint planning and management
6. Product backlog management
7. Sprint backlog management
8. Workflow/status management
9. Priority management

10. Kanban board visualization
 11. Task dependencies handling
 12. Time tracking for tasks
 13. Progress reporting
 14. Team performance analytics
 15. File attachments for tasks
 16. Task commenting and discussion
 17. Notification alerts
 18. Activity history tracking
 19. Search and filtering of tasks/projects
 20. Project dashboard overview
-

2.12 Software Development Methodology

The project will follow the **Agile Scrum methodology**, suitable for iterative development and continuous feedback.

Development will occur in short sprints, with regular reviews and incremental feature implementation. This approach supports flexibility, early testing, and progress monitoring.

2.13 Proposed Infrastructure Platform and Tools

- Next.js / React
 - Node.js / Express
 - MongoDB or MySQL
 - Git & GitHub
 - Visual Studio Code
 - Postman
 - Figma / Balsamiq
-

2.14 Project Timeline

Planned phases:

1. Requirements Analysis
2. System Design
3. Implementation

4. Testing
5. Deployment
6. Documentation

(Represented using a Gantt chart in the final document.)

2.15 Mockups

Mockups will include:

- Dashboard screen
 - Project management screen
 - Task board (Kanban)
 - Sprint planning interface
-

2.16 References

[1] Atlassian, “Jira Software,” Available: <https://www.atlassian.com/software/jira>

[2] Atlassian, “Trello,” Available: <https://trello.com>

[3] Asana Inc., “Asana Project Management,” Available: <https://asana.com>

[4] K. Schwaber and J. Sutherland, “The Scrum Guide,” Available: <https://scrumguides.org>